

Cloud Orchestration with Infrastructure as Code (IaC): The Benefits

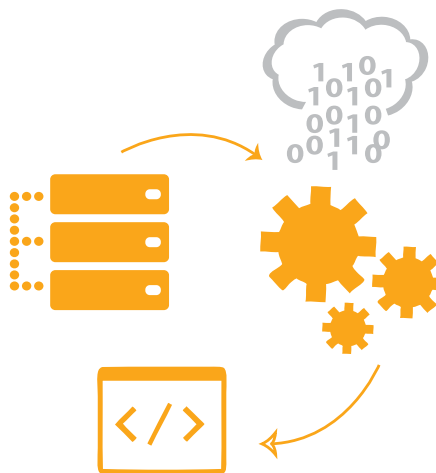
IaC is the art of defining infrastructure using code, making it repeatable and predictable. With cloud orchestration, you can deploy and manage your infrastructure in a more agile and efficient way. No more manual configurations or guesswork!

What is it that makes Infrastructure as Code (IaC) one of the greatest technologies in the cloud scenario of the 21st century? Well, it is IaC that powers cloud infrastructure. Supported by various highly innovative features and tools, IaC automates IT infrastructure, empowering organisations to quickly build, deploy, and scale cost-efficient cloud applications.

IaC: A blessing for cloud optimisation?

Infrastructure encompasses various components required to deploy software in a cloud environment, such as VMs, containers, serverless functions, security features like IAM (identity and access management) and KMS (key management service), networking, and monitoring and logging capabilities.

Now, let's look at the Google Play Store. You download tens to hundreds of apps to your Android device from there. In the past, updating these apps required manual effort. You had to individually check for updates and install them. However, with IaC in place, the process becomes seamless and automated. Through well-defined code and configuration files, the infrastructure behind the Play Store can automatically detect app updates and



efficiently roll out the new versions to users' devices. This ensures that users have the latest features and bug fixes without the need for manual intervention. Thanks to IaC, your apps are now automatically updated.

Virtualisation and cloud development eliminate the challenges of physical hardware management, allowing developers to build their own virtual servers or containers.

As technology rapidly evolves, it is important for IT infrastructure development staff to set up and update the cloud environment from time to time. Data administrators need to determine the resources required for coding, ranging from network configuration to data storage in cloud environments.

IaC: A force to be reckoned with in cloud orchestration

IaC is playing an important role in IT supporting frameworks like servers, hard disk drives, etc; applying security updates, installing services and applications, establishing networks, and managing memory. It overcomes common IT problems such as configuration drifting, operator mistakes, inefficiencies and information loss. Flexible licensing models and reduced maintenance costs are a few factors contributing to the dominance of IaC in cloud orchestration.

With evolving consumer behaviour and demands, enterprises must act to stay competitive in the market. Consequently, cloud infrastructures are expanding and IaC is the only method that can handle the complexities associated with them.

Large-scale organisations, such as those in telecommunications, retail, and manufacturing sectors, often struggle to meet customer demands due to work delays. To increase work speed and minimise human errors, automation of cloud infrastructure becomes essential for such organisations. Hence, they are increasingly opting for IaC.

The IaC market size is projected to grow from US\$ 0.8 billion in 2022 to US\$ 2.3 billion in 2027, at a CAGR of 24% during the forecast period.